TrES-3b

R-1601

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_191014` · created 2026-08-02T02:18:36+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458770.6267 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.021
Transit depth (Rp/Rs)^2	3.58 +/- 0.79 [%]
Orbital Inclination (inc)	82.47 +/- 1.33
Airmass coefficient 1 (a1)	1.004 +/- 0.014
Airmass coefficient 2 (a2)	-0.0 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.39 %
Best Comparison Star	#2 - [212, 334]
Optimal Method	PSF photometry
Transit Duration (day)	0.061 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.189 ± 0.021 vs 0.16309 (deviation 1.23σ)
Duration fit vs published	0.061 d vs 0.0567 d (deviation 0.36σ)
Mid-time O–C	1.7 min
Depth SNR	9.0
Residual scatter	1.39 %

Light curve

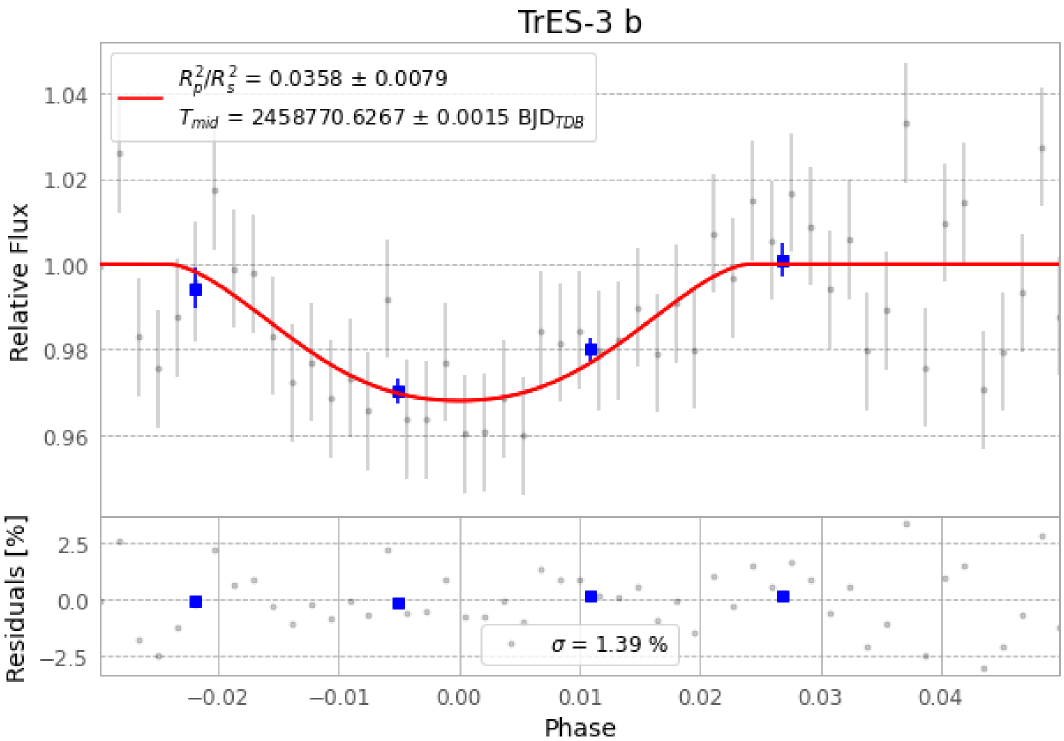

Figure 1 — FinalLightCurve_TrES-3 b_2019-10-13.png (SHA-256 1cc0cb35185e4e6a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 249], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 04f2c41a1a8ac928…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

51 FITS frames (33 MB), `TRES-3191014020614.FITS` → `TRES-3191014043612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-191014.log (36628e709855…), FinalLightCurve_TrES-3 b_2019-10-13.png (1cc0cb35185e…), FinalLightCurve_TrES-3 b_2019-10-13.csv (f120500b82b1…)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)